Interpretable Deep Learning for Whole Genome Analysis

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Thousand Genomes Project (1000G)

Figure 1. The 1000G Populations and Super-Populations. Circles = merged populations

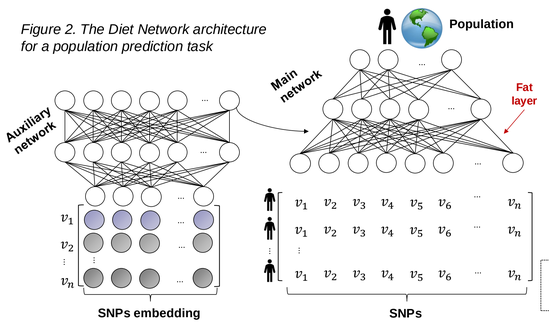
2322 samples from individuals across the world:
- 24 populations (Figure 1);
- 5 super-populations

229,986 Single Nucleotide Polymorphisms (SNPs) called from whole genome sequencing.

Fat data for machine learning: EASY TO OVERFIT!

DietNetwork Architecture

Figure 2. The Diet Network architecture for a population prediction task



Main Network: MLP that predicts genetic ancestry given genotypes (Fig. 2)

Auxiliary Network: using SNP embeddings to constrain network weights in the 230k SNPs x 100 "fat layer"
SNPs embedding = Population Genotype Frequencies

REDUCES OVERFITTING!
HARD TO INTERPRET!

Input Dropout

We experimented with an additional hyperparameter: dropping out the inputs at a given rate. We found that this improved the generalizability to CARTaGENE.

The ideal amount of input dropout appears to be 0.94. Interestingly, this is:

$$\approx 1 - \frac{Training\ set\ size}{Number\ of\ features}$$

I.e. the number of non-masked SNPs is equal to the size of the training set.

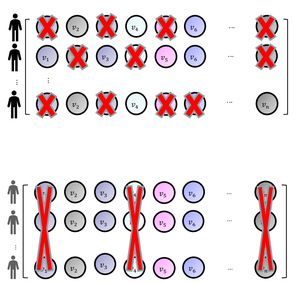


Figure 3. Input dropout scheme

Generalizability

Generalization to other datasets imply to handle missing data (Figure 3). We trained the DietNetwork on 1000G with different rates of input-dropout which provides improvement of the accuracy (Figure 4).
In order to test the generalization capabilities, we show here the predicted class of self-declared Europeans from the CARTaGENE biobank (CaG), with different rate of missing data (Figure 5).

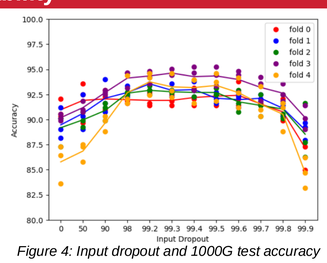


Figure 4: Input dropout and 1000G test accuracy

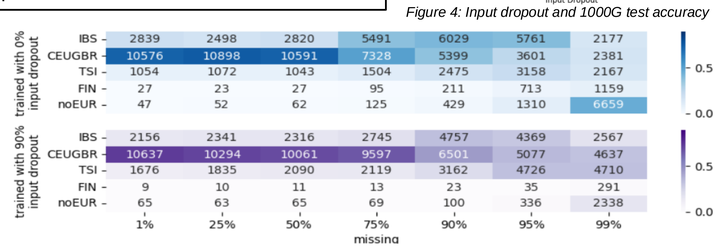


Figure 5: Input dropout and generalizability on CaG

Interpretability

Integrated Gradients: This approach calculates an attribution for each feature of each sample. The attribution is determined by summing the gradients of the DietNetwork output for some synthetic samples. These samples are generated from equidistant interpolations between the input and a designated *baseline* (Figure 6).

UNIQUELY SATISFIES A SET OF AXIOMS AND ARE FAST TO COMPUTE!
HARD TO EVALUATE, NO GROUND TRUTH!

Baseline: An input intended to represent the 'absence of information' and produce a 'neutral prediction'. Alternatively, we could average from multiple baselines drawn from a distribution.

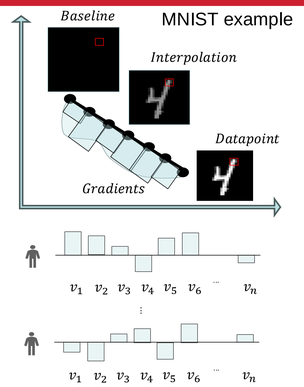


Figure 6. Integrated Gradient Attribution Method

Research Question and Aims

Research Question: How do we best leverage Integrated Gradient attributions to improve the interpretability of a deep learning network on fat genomics data?
Aims:
1. Set a framework to compare performance of Integrated Gradients attributions in different contexts.
2. Evaluate different baseline accurate for the genotype modality
3. Understand the attributions by comparing them to existing population genomics statistics

Method

Single Baseline	Multiple Baselines
• **Reference Genome** • **Missing Genome** • **Most Uniform Sample**	**Random Sample** **Random Generated:** • **Uniform Genotype Frequency** • **Global Frequency** • **Per-population Frequency**

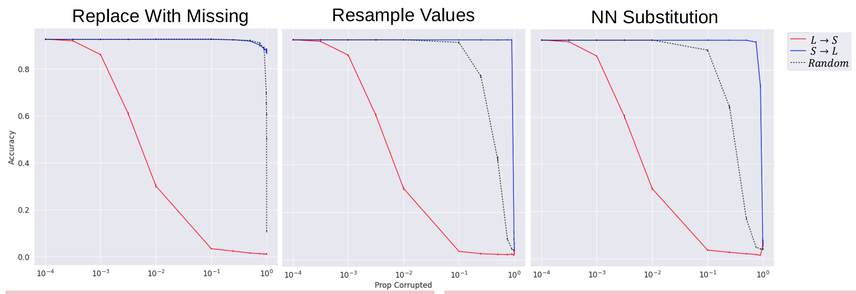
Attribution Performance Metric

- When we corrupt a set of SNPs, the accuracy decreases.
- The greater the decrease, the more significant the SNPs were in the prediction task.
- We increase the % corrupted from ~0 to ~100, recomputing accuracy on the test set each time.
- We compute area under the curve (Figure 7)

Corruption by:
- Replacing with baseline value
- Resampling the value
- Substitution with Nearest Neighbours value

Corrupt:
- $AUC\ L \rightarrow S$: largest attributed SNPs to smallest
- $AUC\ S \rightarrow L$: Smallest to largest
- Random Set of SNPs (Control)

Figure 7. We report the area under these curves. For visualization, we show the curves of the missing baseline.



Population Differentiation Scores (F_{ST})

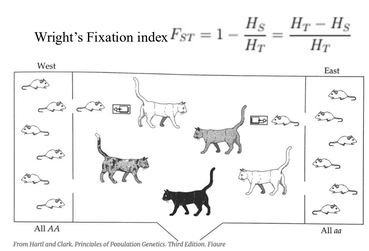


Figure 8. Extreme FST example

High F_{ST} score = SNP is much less heterozygous in subpopulations vs the large population
⇒ SNPs are differentiated between populations

Attribution Matrix

For each SNP, we computed the average attribution for all the members of each population (row) with respect to each population (column). See figure 9.

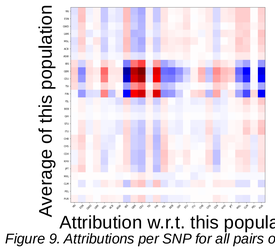


Figure 9. Attributions per SNP for all pairs of populations

We compared this to the matrices of F_{ST} values.

Results

Baseline Comparisons

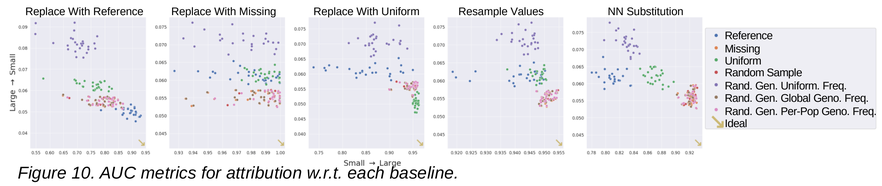


Figure 10. AUC metrics for attribution w.r.t. each baseline.

Table 1. Raw data for Figure 10. We also display random corruption AUC

Corruption Style	Reference AUC L→ S	Reference AUC S→ L	Missing AUC L→ S	Missing AUC S→ L	Uniform AUC L→ S	Uniform AUC S→ L	Random SNP Corruption
Replace with Missing	0.0613 ± 0.0016	0.9079 ± 0.0158	0.0553 ± 0.0014	0.9109 ± 0.0139	0.0614 ± 0.0014	0.9170 ± 0.0099	0.9022 ± 0.0092
Replace with Reference	0.0497 ± 0.0019	0.8285 ± 0.0270	0.0549 ± 0.0016	0.7389 ± 0.0410	0.0624 ± 0.0018	0.7031 ± 0.0474	0.4148 ± 0.0413
Replace with Uniform	0.0611 ± 0.0016	0.7924 ± 0.0515	0.0557 ± 0.0012	0.8764 ± 0.0140	0.0504 ± 0.0017	0.8855 ± 0.0092	0.4118 ± 0.0162
Resample Value	0.0613 ± 0.0015	0.8717 ± 0.0112	0.0553 ± 0.0014	0.8801 ± 0.0082	0.0615 ± 0.0014	0.8777 ± 0.0082	0.4463 ± 0.0178
1NN Substitution	0.0618 ± 0.0020	0.7512 ± 0.0158	0.0559 ± 0.0015	0.8538 ± 0.0106	0.0622 ± 0.0020	0.8070 ± 0.0118	0.3450 ± 0.0110

Corruption Style	Random Sample AUC L→ S	Random Sample AUC S→ L	Random Gen Sample: Uniform Genotype Freq. AUC L→ S	Uniform Genotype Freq. AUC S→ L	Global Genotype Freq. AUC L→ S	Global Genotype Freq. AUC S→ L	Per Pop Genotype Freq. AUC L→ S	Per Pop Genotype Freq. AUC S→ L
Replace with Missing	0.0554 ± 0.0012	0.9192 ± 0.0100	0.0713 ± 0.0023	0.9041 ± 0.0169	0.0554 ± 0.0014	0.9112 ± 0.0138	0.0554 ± 0.0013	0.9193 ± 0.0096
Replace with Reference	0.0547 ± 0.0015	0.7522 ± 0.0402	0.0823 ± 0.0042	0.6504 ± 0.0520	0.0550 ± 0.0016	0.7388 ± 0.0408	0.0547 ± 0.0015	0.7515 ± 0.0412
Replace with Uniform	0.0558 ± 0.0010	0.8761 ± 0.0152	0.0707 ± 0.0026	0.8131 ± 0.0254	0.0557 ± 0.0011	0.8761 ± 0.0140	0.0558 ± 0.0010	0.8770 ± 0.0151
Resample Value	0.0554 ± 0.0012	0.8799 ± 0.0085	0.0713 ± 0.0022	0.8741 ± 0.0077	0.0554 ± 0.0013	0.8801 ± 0.0082	0.0554 ± 0.0012	0.8799 ± 0.0085
1NN Substitution	0.0560 ± 0.0015	0.8546 ± 0.0114	0.0720 ± 0.0026	0.7701 ± 0.0145	0.0560 ± 0.0015	0.8530 ± 0.0107	0.0559 ± 0.0015	0.8543 ± 0.0114

- For each corruption method, *For each, we compared 5 folds and 3 random seeds.*
- the AUC for random SNP corruption was larger than $AUC\ L \rightarrow S$, and smaller than $AUC\ S \rightarrow L$ (Fig. 10, with values ranging between 0.345-0.902 vs 0.0497-0.0720 vs 0.7031-0.9193).
- We found that each baseline performed best on the corruption style that replaced values using the baseline.
- Aside from these, we found that the missing baseline had the lowest $AUC\ L \rightarrow S$ and highest $AUC\ S \rightarrow L$, or was tied with the randomly generated baselines (global or per-pop genotype frequencies).
- Given that the missing baseline uses significantly less compute, we can consider it to have performed the best overall (Table 1).

Comparisons

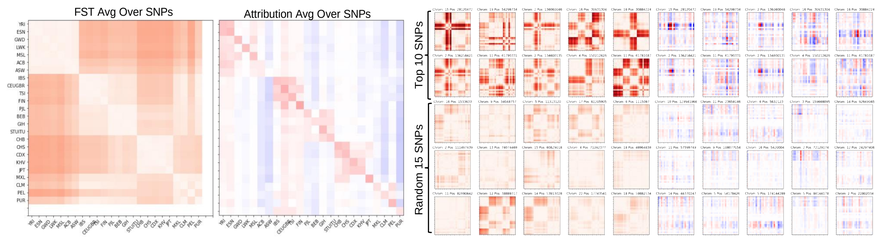


Figure 11. average attribution matrix vs F_{ST} scores. (left) average over SNPs (right) for highest attributed SNPs

For each SNP, we averaged the attributions to get a global importance score. We used these to extract the highest-ranking SNPs in the dataset (Figure 11).

- When we average the F_{ST} scores across all SNPs, we see a block structure due to the super-population structure.
- Likewise, when we look at our attributions, we see members of a super-population more similarly predict populations within their super-population versus the other populations.
- When looking at the per-SNP matrices, we see a resemblance between the F_{ST} and attribution matrices. When a SNP is not differentiated between the individual population and a similar target population, the model will use the SNP as evidence for that target population, but when it is differentiated and the population is distant, the model will use it as evidence against the population.
- This resemblance is strongest for the top attributed SNPs (using attribution averaged across individuals).

Future Directions

Do the models with higher input dropout use different SNPs? We can use interpretability for this!
How do other interpretability methods like SHAP perform?

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